

LOURENZ O. BALIBER, MSc

Researcher | Biophysics | Data Scientist

lbaliber0828@gmail.com • +639323319800 • github.com/loubaliber • <https://loubaliber.github.io>

Physics postgraduate and data scientist with expertise in soft matter and polymers, biophysics, end-to-end machine learning pipelines, predictive modeling, and data-driven application deployment. Proficient in Python, R, and MATLAB with hands-on experience in data analysis, computer vision, and scientific computing. Currently a data science intern at OpsWerks, with a track record of collaborating with international research teams.

TECHNICAL SKILLS

- Programming & Data Tools: **Python (NumPy, Pandas, SciPy, Matplotlib, Scikit-learn, PyTorch), R, MATLAB, Mathematica, C++, Git/Version Control, Linux/Unix & Shell Scripting**
- Data Science & ML: **Supervised & unsupervised learning, regression and classification, time-series analysis, statistical modeling, computer vision (OpenCV), feature engineering, cross-validation & model evaluation, forecasting, NLP, HuggingFace**
- Deployment & Web: **Flask, Streamlit, React, REST APIs, end-to-end ML deployment, SQL, MongoDB, automated data pipelines (n8n, Zapier), JavaScript/HTML/CSS, Docker**
- Scientific Computing: **Molecular dynamics simulation (LAMMPS, GROMACS), physics-based simulation, passive microrheology, particle tracking, fractal dimension analysis**
- Other: **Tableau, LaTeX, scientific writing, CAD modeling**

EXPERIENCE

Data Science Intern

Aug 2026 – Present

OpsWerks, Cebu City, Philippines

- Selected for a **12-week full-time data science program** spanning foundations, intermediate machine learning, and a mentor-approved capstone project.
- Applying Python-based ML workflows to structured datasets through individual and group deliverables, with progress reviewed in weekly 1:1 and team check-ins.
- Maintaining the required **80% average** across all technical assessments while contributing to technical discovery and operations, process design, and refinement.

Exchange Student Researcher (Physical Chemistry)

2025 – 2026

Tokyo University of Marine Science and Technology

- Processed and analyzed **2,000+ fluorescence microscopy image frames** using custom Mathematica pipelines, reducing manual analysis time by **~60%**.
- Developed image analysis workflows correlating structural heterogeneity with particle dynamics from passive Brownian motion tracking across **40+ samples**.
- Applied statistical modeling to characterize phase-separated biopolymer network formation; results and data analysis contributed to a **peer-reviewed publication**.

Graduate Researcher – MSc Thesis

2024 – 2026

Medical Biophysics Group, University of San Carlos

- Performed quantitative analysis of fluorescence microscopy and passive particle tracking on mixed carrageenan hydrogel systems.
- Implemented data processing pipelines in Python/MATLAB for MSD, viscoelastic moduli extraction, and gel network characterization.
- Collaborated with international researchers on biophysics and data analysis methodology.

Research Assistant

Aug – Nov 2023

DOST PCIEERD NICER Project — 3D-BTM: Bio-fabrication of Bone, Muscle & Pancreatic Tissue Models for Printable and Injectable Scaffolds

- Supported data collection and analysis for biomimetic tissue engineering experiments, processing **15+ experimental datasets**.
- Assisted in processing and interpreting biomechanical measurement datasets; contributed to a **government-funded national research project**.

Research Intern

Jun – Jul 2023

Medical Biophysics Group, University of San Carlos

- Diagnosed optical tweezers and microrheology experiments on complex fluids.
- Processed experimental data to quantify local viscoelastic properties of biological samples, achieving reproducibility within **±3% measurement error**.

EDUCATION

MSc – Physics

2024 – 2026

University of San Carlos, Cebu City, Philippines

- Thesis: "Gelation Mechanism and Network Formation of FITC-Labeled Mixed Kappa/Lambda Carrageenan Gels"
- DOST ASTHRDP Scholar

BSc – Applied Physics

2020 – 2024

University of San Carlos, Cebu City, Philippines

- Thesis: "Characterization of Fibrin-Carrageenan Mixtures as a Hydrogel Model Using Passive Particle Tracking and AFM for Skeletal Muscle Tissue Engineering"
- DOST JLSS Scholar (2022–2024) | Ironwood Scholar (2020–2024) | Dean's Lister (2020–2022)

PUBLICATIONS

- Baliber, L. et al. (2026). Brownian motion in a phase-separated biopolymer network. *Proceedings of the 44th Samahang Pisika ng Pilipinas Physics Conference*, SPP-2026-PA-01.
- Baliber, L., et al. (2025). Surface roughness and fractal dimension characterization of the topography in pristine and defected nanostructure arrays. *Proceedings of the Samahang Pisika ng Pilipinas*, 43, SPP-2025-2G-02.
- Baliber, L., et al. (2024). Passive microrheology in carrageenan–fibrin mixtures using video microscopy. *Proceedings of the Samahang Pisika ng Pilipinas*, 42, SPP-2024-1E-05.

PROJECTS

Coarse-Grained Polymer Simulation — Molecular Dynamics

2026

LAMMPS · Python · OVITO | github.com/loubaliber/cg-polymer-md

- Built coarse-grained molecular dynamics simulations of polymer chains in **LAMMPS**, with Python-driven system setup, trajectory post-processing, and OVITO visualization of chain conformation and network structure.

Signal Translator — Audio Fingerprinting Desktop App

2026

Python · FFT · MFCC | github.com/loubaliber/signal-translator

- Captures short vocal signals through the microphone and fingerprints them using **FFT and MFCC audio features**, matching new input against a library of user-taught signals so custom "words" can be learned and recognized later.

Hands Detection Music — Real-Time Computer Vision Instrument

2026

Python · OpenCV | github.com/loubaliber/hands-detection-music

- Real-time webcam hand tracking mapped to live audio synthesis: the left hand selects the chord and the right hand controls how it is played, running end-to-end on a standard laptop camera feed.

Climate Delhi — Time-Series EDA & Regression

2026

Python · Pandas · Scikit-learn | github.com/loubaliber/climate-delhi

- Exploratory analysis and regression modeling notebook predicting daily mean temperature in Delhi from the **2013–2017 Daily Climate Time Series** dataset, covering feature engineering, seasonality, and model evaluation.

Stroke Detection System — End-to-End ML Pipeline

2024

Flask · Streamlit · Scikit-learn | github.com/fitaness12345/Stroke-Dataset-Prediction

- Built a complete stroke detection system from EDA to web deployment on a dataset of **5,000+ patient records**.
- Deployed as an interactive web application using **Flask + Streamlit**, enabling real-time risk prediction with visual feature importance output.

Satellite Launch & Orbit Simulator — Python Orbital Mechanics Model

2025

Python · RK4 · Matplotlib

- Built a 2D physics simulation modeling a satellite's trajectory from launch to orbital insertion at LEO, solving the equations of motion via **custom RK4 integration** under gravitational and atmospheric drag forces while accounting for fuel consumption and mass loss.
- Visualized trajectory, velocity, and altitude profiles in real time, validating simulation output against analytical orbital mechanics solutions.

Additional Tools

2023 – 2026

- **Balibot** — Discord bot built with discord.py serving **200+ users across 5+ servers**; automated moderation, slash commands, API integrations, **96.3% uptime** over 6 months.
- **latex-hub.nvim** — Neovim plugin (Lua) providing a fast project switcher and status dashboard for LaTeX projects.
- **Gmail Scraper** — Python/IMAP utility that archives an entire Gmail mailbox and its attachments to plain-text files.

HONORS & AWARDS

- 25th SPVM National Physics Conference – Poster Presenter (2023)
- 33rd Annual Fr. Heinrich Schoenig Biosymposium – Poster Presenter (2024)
- ODU REYES Python & Subatomic Physics Mentorship (2021)
- Top 3 Scholar, USC DOST Thesis Grants (2024)
- 44th SPP Conference – Poster Presenter (2026)

AFFILIATIONS

- Creative Writer - The Josenian Premier (2019 - 2020)
- **Administrative Positions**
 - Internal Affairs Councilor - Carolinian Physics Society (2021 - 2022)
 - President – USC Physics and Astronomy Society (2022 - 2023)
 - Councilor - School of Arts and Sciences Council (2022 - 2023)
 - Physics Representative – DOST ASTHRDP Officer (2024-2025)
- **Co-organized Conferences**
 - 2nd USC-VSU Biophysics Symposium (2023)
 - 8th MBG Biophysics Symposium (2024)
 - 3rd USC-VSU Biophysics Symposium (2025)

- 8th International and 16th Annual National Scientific Meeting of the Philippine Society for Cell Biology (2025)
- 18th International Hydrocolloids Conference (2026)